



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

AD2001-1 – FUNDAMENTALS OF MACHINE LEARNING
(AI&DS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Designing a machine learning approach involves**
 - A) Choosing the type of training experience
 - B) Choosing the target function to be learned
 - C) Choosing a representation for the target function
 - D) All of these
2. **Which types of machine learning algorithm uses unlabeled data?**
 - A) Supervised
 - B) Unsupervised
 - C) Instance based
 - D) Regression
3. **What type of machine learning involves learning from labeled data to make predictions or decisions?**
 - A) Unsupervised learning
 - B) Supervised learning
 - C) Reinforcement learning
 - D) Semi-supervised learning
4. **In machine learning, what is the purpose of the testing phase?**
 - A) To evaluate the model's performance on unseen data
 - B) To fine-tune model parameters for better accuracy
 - C) To increase the complexity of the model
 - D) To train the model using labeled data.
5. **The FIND-S Algorithm starts from the most specific hypothesis and generalize it by considering only _____ examples.**
 - A) Negative
 - B) null
 - C) Positive
 - D) Unlabeled data
6. **Which algorithm was introduced to address the drawbacks of the Find-S algorithm?**
 - A) Decision Tree
 - B) K-NN
 - C) K-Means
 - D) Candidate-Elimination algorithm
7. **Which type of linear regression deals with only one predictor variable?**
 - A) Univariate Linear Regression
 - B) Multivariate Linear Regression
 - C) Polynomial Regression
 - D) None of these
8. **What is the purpose of regularized regression in linear regression?**
 - A) Reduce the variance of the model
 - B) Control regression coefficients
 - C) Prevent overfitting
 - D) All of these
9. **What is the primary task of logistic regression?**
 - A) Predict the values of categorical variables
 - B) Used for solving classification problems
 - C) We can find the S-curve
 - D) All these
10. **What is the purpose of the activation function in an artificial neuron?**
 - A) It introduces nonlinearity to the network, enabling it to learn complex patterns.
 - B) It adjusts the weights of connections between neurons.
 - C) It determines the importance of input features.
 - D) It calculates the error between predicted and actual outputs.
11. **In SVM, what are support vectors?**
 - A) Hyperparameters controlling the model's complexity
 - B) Feature set
 - C) Data points located close to the decision boundary
 - D) Output values of the classifier

12. The perceptron rule can fail to converge if the examples are _____.
 A) linearly separable B) Not linearly separable
 C) Vector of real-valued D) None of these
13. The instance-based learner is also called
 A) Lazy-learner B) Fast learner
 C) Eager learner D) Old learner
14. Which of the following statement is TRUE about the Bayes classifier?
 A) Bayes classifier works on the Bayes theorem of probability B) Bayes classifier is an unsupervised learning algorithm
 C) Bayes classifier is also known as maximum a priori classifier D) It assumes the independence between the independent variables or features
15. Decision tree learning is a method for approximating _____ target functions, in which the learned function is represented by a decision tree.
 A) Continuous-valued B) Discrete-valued
 C) Boolean-valued D) None of these
16. What is the purpose of padding in convolutional neural networks?
 A) To increase the receptive field of the neurons B) To reduce computational complexity
 C) To prevent the loss of information at the edges of the input D) To introduce non-linearity into the network
17. Which algorithm is frequently employed to divide data into clusters by minimizing the sum of squares within each cluster?
 A) K-NN B) K-Means
 C) SVM D) Naïve Bayes
18. What distinguishes reinforcement learning from other types of machine learning?
 A) It solely relies on unsupervised learning algorithms B) Relying solely on labeled data
 C) It involves learning from rewards and punishments D) It is used for classifying the data
19. Which evaluation measure is used to assess the performance of classification algorithms?
 A) Accuracy B) Precision
 C) Recall D) All these
20. What does Q-learning primarily focus on?
 A) Discovering optimal policies B) Minimizing the total number of actions
 C) Minimizing turnaround time D) Balancing exploration and exploitation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

- | Unit – I | | Marks | BT* | CO* | PO* |
|----------|--|-------|-----|-----|-----|
| 1. | a) What is machine learning? Give a detailed explanation of any seven applications of machine learning. | 8 | L1 | 1 | 1 |
| | b) Explain the differences between supervised, unsupervised, and reinforcement learning. Give an example scenario for each type. | 8 | L2 | 1 | 1 |
| 2. | a) Define error and noise in the context of machine learning. How do these factors affect the performance of a learning algorithm? | 5 | L1 | 1 | 1 |
| | b) Explain the concepts of bias and variance in machine learning. How do these factors influence the performance and generalizability of a model? | 5 | L2 | 1 | 1 |
| | c) Apply candidate elimination algorithm to find out hypothesis and final version space for the EnjoySport concept by considering below given training instance. | | | | |

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

6 L3 2 2

3. a) What is regression? Explain different types of linear regression. 8 L2 2 1
 b) Illustrate the process of gradient descent with a detailed example in the context of optimizing a linear regression model. 8 L2 2 1

Unit – II

4. a) Compare and contrast regression and classification algorithms. 5 L2 3 1
 b) Explain the concept of a perceptron with a neat diagram. 5 L1 3 1
 c) Discuss how a classification problem can be solved using SVM. 6 L3 3 1
5. a) Illustrate the stochastic gradient descent version of the back propagation algorithm for feedforward networks containing two layers of sigmoid units. 8 L2 3 1
 b) Apply the decision tree algorithm and construct the tree using following dataset:

Day	Out look	Temperature	Humidity	Wind	Play tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

6. a) Discuss the K-nearest neighbor (KNN) model and its approach to classification with an example. 8 L3 4 2
 b) Explain Bayes theorem. Apply naïve bayes classifier algorithm for learning and classifying text. 8 L2 4 1
 8 L3 4 1

Unit – III

7. a) Explain the K-means clustering algorithm and provide an illustrative example to demonstrate its operation. Evaluate its strengths and limitations. 8 L2 5 1
 b) What is Reinforcement Learning? Explain the Reinforcement learning with a neat diagram. 8 L2 5 1
8. a) Explain following terms:
 i. Hierarchical clustering
 ii. Cross Validation
 iii. Measuring Error 8 L2 5 2
 b) Describe the Q-learning algorithm and its significance in reinforcement learning. Discuss its key components and how it enables an agent to learn optimal policies. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
